

Connection and Selection of Large Current Split-Core Transformer





The function of the current transformer (CT for short) is to convert the primary current with a large value into a secondary current with a small value through a certain transformation ratio, which is used for protection, measurement and other purposes. According to the CT ratio, the reading will be converted into an analog signal, which can be used by the power calculation equipment to obtain the actual current value.

In the actual use of photovoltaic power generation, the current may be large when the inverters are paralleled, and a large current CT is required for sampling. Taking the ESCT-B816 series as an example, this article will introduce the wire selection, installation method and specification selection of high-current CT.

Typical Specification (ESCT-B Series)

Specification	
Frequency	50Hz-60Hz
Rated current	100A to 6000A loads
Rated output	5A / 1A(AC)
Accuracy	Class 0.5 or 1 from 20% to 120% of rated current
Phase angle	Less than 2 degrees at 50% of rated current
Insulation voltage	600Vac
Maximum primary voltage	5000Vac(Insulated Conductor)
Dielectric strength	2.5KV / 1Ma / 1min
Operating temperature	-15°C to 60°C
Operating humidity	< 85%
Case material	PC / UL 94-V0
Bobbin	PBT
Internal structure	Epoxy

CT selection

On the Meter specification, you will find the current parameter like 1.5 (6) A, that is, the rated current is 1.5A, the maximum current is 6A, and 5A is often used. In general, CT guarantees the rated accuracy within a certain power range, for example, within 20%~120% of the rated current, the Class 0.5 accuracy is guaranteed (see the CT specification for the specific actual situation). The transformation ratio of the current transformer refers to the ratio of the primary side current to the secondary side current, which is set on the instrument screen according to the CT model.

Take the ESCT-B816 1500A/5A CT as the example, the secondary side current is 5A, the effective range of accuracy is 300A~1800A, and the transformation ratio is 300.

If the CT provided by SolaX cannot meet the current requirements, you can choose a third-party CT on the market as the below table shown. Please note SolaX can not guarantee the performance if you are using third-party CT.

Ratio(A)	Burden(VA)		Line section (mm ²)	Line length (m)
	Class:0.5	Class:1.0		
1500A/5A	7.5	-	$2.5\text{mm}^2 \leq \text{CT wire} \leq \text{The requirements of the meter}$ or $13\text{AWG} \geq \text{CT wire} \geq \text{The requirements of the meter}$	< 2.5
2000A/5A	10	-		
3000A/5A	20	-		
4000A/5A	20	-		
5000A/5A	20	-		
6000A/5A	20	-		

1500A/5A: Ratio of primary side current to secondary side current.

Class 0.5: The accuracy level of the transformer is 0.5, the higher the level, the lower the accuracy.

Burden(VA): The load on the secondary side of the current transformer. When the actual secondary side load is not greater than the rated load, the selected accuracy can be guaranteed.

Please refer to the measurement range and accuracy requirements of the actual project to select the ammeter and CT.

Wire section

The cross section of wire that connected CT and meter should be no less than 2.5mm^2 , also you need follow the requirement of the meter (you can check this on the smart meter manual). The copper core wire is recommended

$$2.5\text{mm}^2 \leq \text{CT wire} \leq \text{The requirements of the meter}$$

$$\text{or } 13\text{AWG} \geq \text{CT wire} \geq \text{The requirements of the meter}$$

Wire length

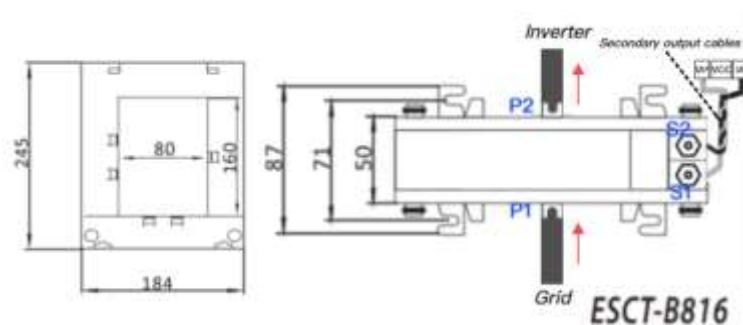
The longest secondary line (the wire from CT to meter) of the current transformer is determined by the capacity of the current transformer, the secondary rated current, the section of the line and the size of the load. According to the impedance and cross-sectional area of the line, combined with the connection between the CT and the meter in the actual site, the length of the secondary side line is recommended to be less than 2.5m.

Shielding requirements

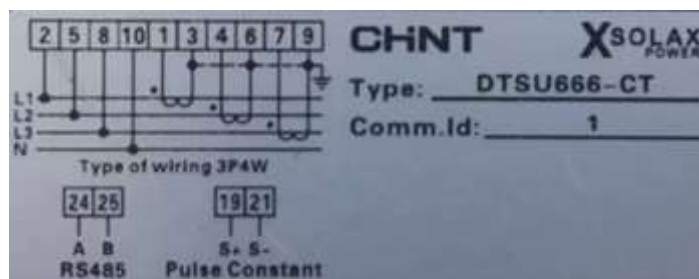
Signal level and shielding, when the signal level is small, it is susceptible to interference from noise signals, and shielded wires are often used to overcome interference. The function is to prevent external interference sources from entering the signal source.

It is necessary to use shielded wires if the secondary current is less than 1A. In our application the secondary current normally is 5A which does not need shielding wire.

Connection



- S1 must be connected to the incoming line of the corresponding phase ammeter, and S2 must be connected to the outgoing line. Take the DTSU666-CT as example, connect S1 to pin1 and connect S2 to pin3 for L1



- P1 is the incoming side of the primary line (usually the Grid side), P2 is the outgoing side of the primary line (usually the Inverter side).
- For the sake of safety, the S2 terminal of the transformer must be grounded with a copper wire of more than 2.5 square meters.